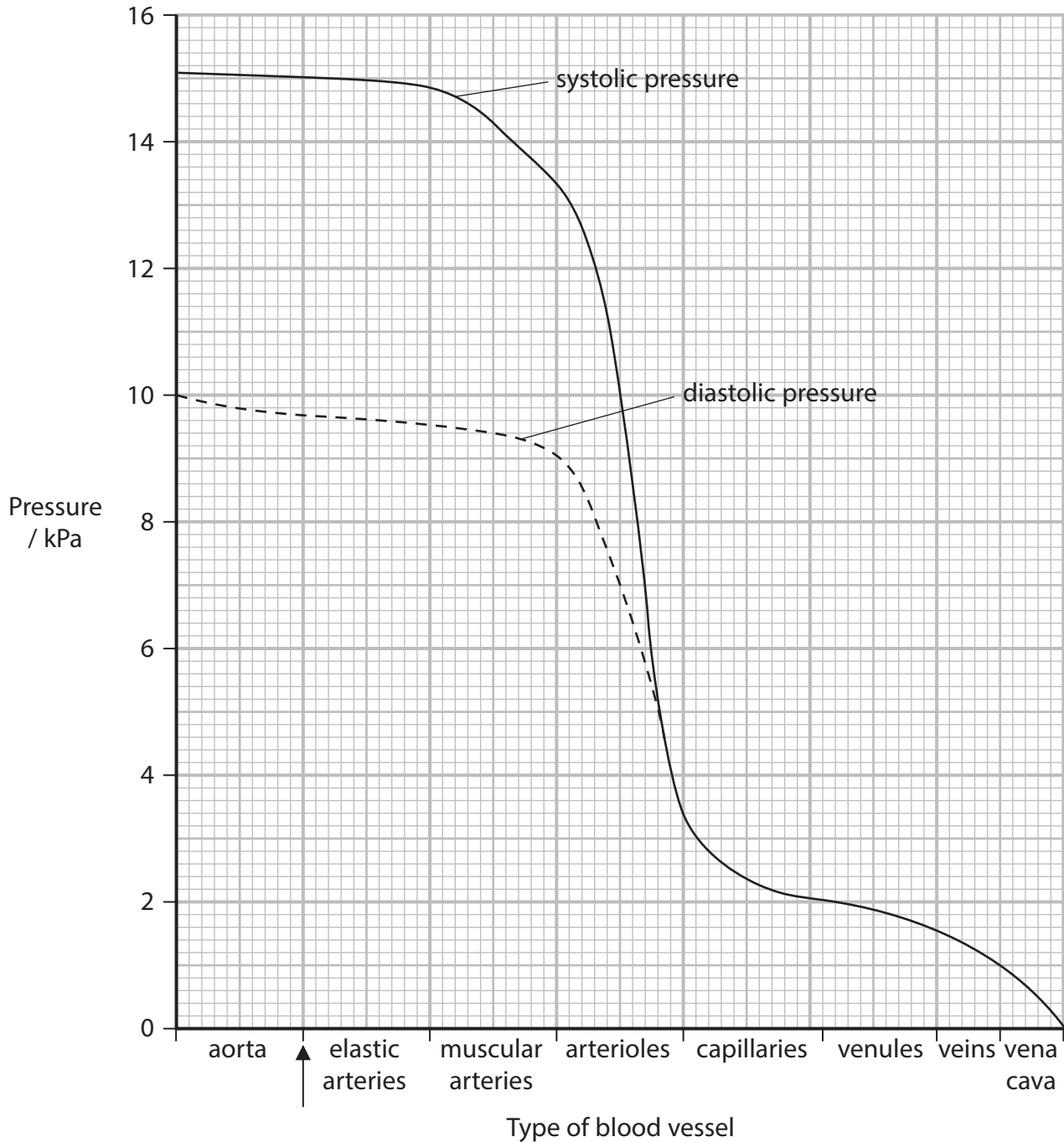


7 Blood flow and blood pressure are affected by a number of factors.

Abnormal blood flow and abnormal blood pressure affect the health of a person.

- (a) The graph shows the systolic pressure and the diastolic pressure as blood flows through human blood vessels.



- (i) Pulse pressure is the difference between systolic pressure and diastolic pressure.
- In a healthy person, pulse pressure should be at least 25% of the systolic pressure.
- Determine whether this person is healthy, using the point indicated by the arrow on the x-axis of the graph.

(2)

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- (ii) The mean arterial pressure (MAP) can be approximated using the formula

$$\text{MAP} = \text{diastolic blood pressure} + \frac{(\text{systolic blood pressure} - \text{diastolic blood pressure})}{3}$$

Estimate the MAP in the elastic arteries for this person.

(2)

Answer kPa



- (iii) If the mean arterial pressure (MAP) value is less than 8 kPa, the force pushing the blood through the vessels will be too low.

Explain how a low MAP could affect a person.

(2)

- (b) Blood flow depends on the change in pressure (ΔP), the radius of the blood vessel lumen (r), the length of the blood vessel (λ) and the viscosity of the blood (η).

Blood flow can be calculated using the formula

$$\text{Blood flow} = \frac{\pi \Delta P r^4}{8 \eta \lambda}$$

Blood flow can also be calculated using the formula

$$\text{Blood flow} = \frac{\Delta P}{\text{resistance}}$$



(i) Which formula can be used to calculate resistance?

(1)

☐ **A** $\frac{8\eta\lambda}{\pi\Delta Pr^4}$

☐ **B** $\frac{\pi\Delta Pr^4}{8\eta\lambda}$

☐ **C** $\frac{\pi r^4}{8\eta\lambda}$

☐ **D** $\frac{8\eta\lambda}{\pi r^4}$

(ii) A change in which factor has the biggest effect on resistance?

(1)

☐ **A** blood pressure

☐ **B** length of the blood vessel

☐ **C** radius of the blood vessel lumen

☐ **D** thickness of the blood vessel wall

(iii) Explain how arteries are adapted to reduce resistance to blood flow.

(2)

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- (c) Blood vessels can expand to accommodate increased blood flow. This is known as compliance.

- (i) Explain how arteries are adapted to accommodate sudden increases in blood flow.

(2)

- (ii) Explain why a reduction in the compliance of a blood vessel can lead to cardiovascular disease (CVD).

(3)

[illegible]

(Total for Question 7 = 15 marks)

